

Question Paper: Machine Learning

****Machine Learning Question Paper****

****Instructions:****

- * The question paper consists of three sections: Multiple Choice Questions (MCQs), Short Questions, and Long/Descriptive Questions.
- * Attempt all questions in the respective sections.
- * Use a blue or black pen to write your answers.

****Section A: Multiple Choice Questions (30 marks)****

Choose the correct option for each question:

1. What is the primary goal of Machine Learning?

- A) To develop software that can reason like humans
- B) To develop software that can learn from data
- C) To develop software that can only perform mathematical calculations
- D) To develop software that can only interact with humans

Correct answer: B) To develop software that can learn from data

2. Which of the following is a type of supervised learning?

- A) Clustering
- B) Regression

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C) Dimensionality reduction

D) Association rule mining

Correct answer: B) Regression

3. What is the term for the process of reducing the dimensionality of a dataset?

A) Feature extraction

B) Feature selection

C) Dimensionality reduction

D) Data augmentation

Correct answer: C) Dimensionality reduction

4. Which algorithm is commonly used for image classification tasks?

A) Linear Regression

B) Decision Trees

C) Convolutional Neural Networks (CNNs)

D) Support Vector Machines (SVMs)

Correct answer: C) Convolutional Neural Networks (CNNs)

5. What is the purpose of regularization in Machine Learning?

A) To increase the complexity of a model

B) To decrease the complexity of a model

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C) To increase the training time of a model

D) To decrease the testing time of a model

Correct answer: B) To decrease the complexity of a model

6. Which of the following is a type of evaluation metric for classification models?

A) Mean Squared Error (MSE)

B) Mean Absolute Error (MAE)

C) Accuracy

D) Precision

Correct answer: C) Accuracy

7. What is the term for the process of training a model on a small dataset and then fine-tuning it on a larger dataset?

A) Transfer learning

B) Active learning

C) Semi-supervised learning

D) Reinforcement learning

Correct answer: A) Transfer learning

8. Which of the following is a popular library for Machine Learning in Python?

A) NumPy

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- B) pandas
- C) scikit-learn
- D) TensorFlow

Correct answer: C) scikit-learn

9. What is the purpose of data preprocessing in Machine Learning?

- A) To increase the size of a dataset
- B) To decrease the size of a dataset
- C) To improve the quality of a dataset
- D) To reduce the complexity of a model

Correct answer: C) To improve the quality of a dataset

10. Which of the following is a type of neural network architecture?

- A) Decision Trees
- B) Random Forest
- C) Convolutional Neural Networks (CNNs)
- D) Support Vector Machines (SVMs)

Correct answer: C) Convolutional Neural Networks (CNNs)

****Section B: Short Questions (30 marks)****

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Answer each question in 3-5 marks:

1. What is the difference between supervised and unsupervised learning? (5 marks)
2. Explain the concept of overfitting in Machine Learning. (4 marks)
3. What is the purpose of feature scaling in Machine Learning? (3 marks)
4. Describe the process of training a neural network. (5 marks)
5. What is the difference between a regression and classification problem? (4 marks)

****Section C: Long/Descriptive Questions (30 marks)****

Answer each question in 8-10 marks:

1. Explain the concept of deep learning and its applications in Machine Learning. (10 marks)
2. Describe the process of building a recommendation system using collaborative filtering. (8 marks)
3. What is the concept of reinforcement learning? Explain its applications and challenges. (10 marks)

****Answer Key:****

****Section A: Multiple Choice Questions****

1. B) To develop software that can learn from data
2. B) Regression
3. C) Dimensionality reduction

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4. C) Convolutional Neural Networks (CNNs)
5. B) To decrease the complexity of a model
6. C) Accuracy
7. A) Transfer learning
8. C) scikit-learn
9. C) To improve the quality of a dataset
10. C) Convolutional Neural Networks (CNNs)

****Section B: Short Questions****

1. Supervised learning involves training a model on labeled data, while unsupervised learning involves training a model on unlabeled data.
2. Overfitting occurs when a model is too complex and performs well on the training data but poorly on the testing data.
3. Feature scaling is used to normalize the features of a dataset to have similar scales.
4. Training a neural network involves feeding the network with input data, calculating the output, and adjusting the weights to minimize the loss function.
5. A regression problem involves predicting a continuous output variable, while a classification problem involves predicting a categorical output variable.

****Section C: Long/Descriptive Questions****

1. Deep learning involves using neural networks with multiple layers to learn complex patterns in data. Applications include image classification, natural language processing, and speech

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recognition.

2. Building a recommendation system using collaborative filtering involves creating a matrix of user-item interactions, calculating the similarity between users, and recommending items to users based on the preferences of similar users.

3. Reinforcement learning involves training an agent to take actions in an environment to maximize a reward signal. Applications include robotics, game playing, and autonomous vehicles. Challenges include exploring the environment, learning from feedback, and avoiding overfitting.